

Machine Translation Hallucination Detection for Low and High Resource Languages using Large Language Models

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Abstract

Recent advancements in massively multilingual machine translation systems have significantly enhanced translation accuracy; however, even the best performing systems still generate hallucinations, severely impacting user trust. Detecting hallucinations in Machine Translation (MT) remains a critical challenge, particularly since existing methods excel with High-Resource Languages (HRLs) but exhibit substantial limitations when applied to Low-Resource Languages (LRLs). This paper evaluates sentence-level hallucination detection approaches using Large Language Models (LLMs) and semantic similarity within massively multilingual embeddings. Our study spans 16 language directions, covering HRLs, LRLs, with diverse scripts. We find that the choice of model is essential for performance. On average, for HRLs, Llama3-70B outperforms the previous state of the art by as much as 0.16 MCC (Matthews Correlation Coefficient). However, for LRLs we observe that Claude Sonnet outperforms other LLMs on average by 0.03 MCC. The key takeaway from our study is that LLMs can achieve performance comparable or even better than previously proposed models, despite not being explicitly trained for any machine translation task. However, their advantage is less significant for LRLs. ¹

1 Introduction

Text generation models have drastically improved in recent years especially with the capabilities of LLMs in producing realistic and fluent output. However, hallucination continues to undermine user trust, as it generates and propagates misinformation and sometimes nonsensical outputs (Agarwal et al., 2018; Xu et al., 2023a; Guerreiro et al., 2023b). This issue is especially critical in high-stakes domains like medicine and law, where hallucinations in medical texts can result in harmful

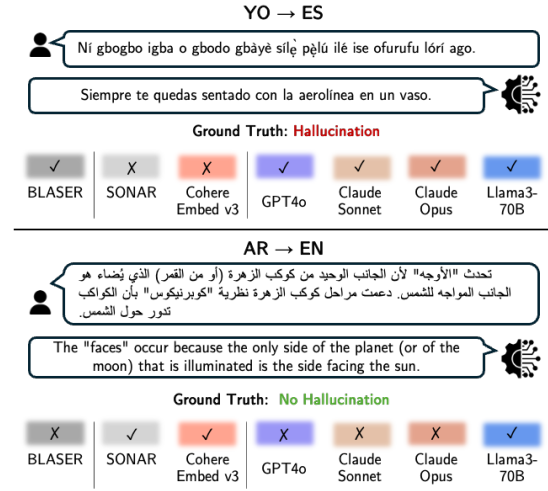


Figure 1: Illustration of how a selection of the evaluated methods perform from Yoruba to Spanish and from Arabic to English.

misunderstandings about diagnoses or treatment instructions, and inaccuracies in legal contract translations can lead to severe financial or legal consequences.

One practical way of reducing hallucination in MT is by building more robust models, especially for LRL which tend to exhibit significantly higher hallucination rates. There are several efforts on scaling MT models to LRLs, such as M2M-100 (Fan et al., 2020), NLLB-200 (Team et al., 2022), MADLAD-400 (Kudugunta et al., 2023) etc. Despite initiatives to minimize hallucinations during the MT process, issues still persists. Therefore, detecting hallucinations post-translation remains a critical alternative approach to ensure the reliability and trustworthiness of the translated content.

Previous work on post-translation evaluation has primarily focused on general translation errors, with evaluation scores often under-representing the impact of hallucinations due to their relatively low frequency compared to less severe errors like omissions (Guerreiro et al., 2023a). Studies on hallucinations have mainly concentrated on English-

¹Data and code are available on [GitHub](#).

centric (EN) to HRL translation direction, while research involving non-English LRLs remains limited (Raunak et al., 2022; Xu et al., 2023b).

For instance, sentence similarity measures between source and translated texts using cross-lingual embeddings, such as LASER (Heffernan et al., 2022) and LaBSE (Feng et al., 2022), have proven effective at identifying severe hallucinations (Dale et al., 2022), though their limitations with LRLs are often overlooked (Dale et al., 2023). Recent studies have highlighted the capabilities of LLMs in multilingual MT evaluation, demonstrating strong performances across various languages, although discrepancies remain for LRLs (Zhu et al., 2023; Xu et al., 2023b). Kocmi and Federmann (2023) demonstrated that, when properly prompted, LLMs can assess the quality of machine-generated translations, achieving state-of-the-art results in system-level evaluation for HRLs. Furthermore, Fernandes et al. (2023) pioneered the use of LLMs for MT tasks in LRLs with the introduction of the *AUTOMQM* prompting technique, though this approach primarily targets broader translation errors rather than focusing on hallucination detection.

Recently, Dale et al. (2023) introduced *HalOmi*—a benchmark dataset for detecting hallucination in MT that includes EN $\leftrightarrow$ HRLs (ten directions) and EN $\leftrightarrow$ LRLs (six directions), as well as two non-English directions HRL $\leftrightarrow$ LRL, including different scripts. BLASER-QE (Communication et al., 2023), the state-of-the-art (SOTA) hallucination detector, is reported as the top performer on the *HalOmi* benchmark. It calculates a translation quality score by evaluating the similarity between encoded source texts and machine-translated texts within the SONAR embedding space (Duquenne et al., 2023).

In this paper, we evaluate the performance of LLMs and embedding based methods as hallucination detectors, aiming to enhance performance in both HRLs and LRLs. To this end, we use the *HalOmi* benchmark dataset with a binary sentence-level hallucination detection approach. For our evaluation, we include 14 methods: eight LLMs with different prompt variations, and four embedding spaces by computing the cosine similarity between source and translated texts.

We find that LLMs are highly effective for hallucination detection across both high and low resource languages, although the optimal model selection depends on specific contexts. For HRLs, on average across directions, the Llama3-70B model

significantly surpasses the previous SOTA method, BLASER-QE, by 16 points. Moreover, embedding-based methods have also demonstrated superior performance over the current SOTA in high resource contexts. However, for LRLs, Claude Sonnet is the best performing model, improving previous methods by a smaller difference. More precisely, LLMs outperformed BLASER-QE in five out of eight LRL translation directions, including the non-English-centric ones.

Finally, our research makes the following primary contributions: First, we evaluate a wide range of LLMs for MT hallucination detection and establish that LLMs, despite not being explicitly trained for the task, are competitive and *greatly* outperform even the previous SOTA for HRLs. Second, large multilingual embedding spaces improve upon previously proposed methods and show that they remain competitive for HRLs, but struggle for LRLs. Third, we establish a new SOTA for 13 of the 16 languages that we evaluate on, including high and low resource languages. Surpassing the previous SOTA, which was explicitly trained for the task, on average by 2 MCC points.

2 Experimental setup

2.1 Quality assessment of the dataset

We evaluated our methods on the *HalOmi* dataset. A first dataset filtration involved selecting only natural translations, without perturbations, as findings from perturbed data may not be applicable to the detection of natural hallucinations (Dale et al., 2023).

The validation and test split was decided based on the translation direction. For the validation set, we selected the two translation directions DE $\leftrightarrow$ EN, which encompasses 301 sentences. This choice was made as extensive resources and established benchmarks are available for this language pair (Guerreiro et al., 2023a), with the expectation that the models would exhibit generalizability to less frequently used language pairs. For the test set, the other 16 pairs were used: more precisely, it includes four pairs with English and a HRL (EN $\leftrightarrow$ AR, EN $\leftrightarrow$ ZH, EN $\leftrightarrow$ RU, EN $\leftrightarrow$ ES), three pairs with English and a LRL (EN $\leftrightarrow$ KS, EN $\leftrightarrow$ MN, and EN $\leftrightarrow$ YO), and one non-English HRL-LRL pair (ES $\leftrightarrow$ YO). The test set includes 2,558 sentence pairs. This test set excludes six sentence pairs that were removed due to sensitive content flagged and filtered out by LLMs. A more detailed description of the dataset is available in Appendix A.

2.2 Hallucination detection setting

We consider two settings: (1) **Severity ranking** introduced by the authors of *HalOmi*. (2) **Binary detection**—a new setting we added due to data imbalance and ease of evaluation.

Severity ranking the classification of hallucinations was based on four severity levels: *No Hallucination*, *Small Hallucination*, *Partial Hallucination*, and *Full Hallucination*. This fine-grained categorization aimed to capture the nuances in the extent and impact of hallucinations on the translated output. We use this setting only as **ablation study** in [Appendix B.](#), both for consistency with the *HalOmi* benchmark, but also to assess the relevance of our binary detection approach.

Binary detection In this setting, all three instances of hallucinations were labelled as *Hallucination*, regardless of their severity. We also change the way the evaluation was done in *HalOmi*, with an appropriate prompt ([Appendix C](#)), and threshold calculation for binary classification for embeddings cosine similarity, see [subsection 2.4](#). The primary reason for choosing this setting is the significant class imbalance in *HalOmi*, largely due to the scarcity of hallucinations across different severity levels. Some translation directions have particularly imbalanced data, for example EN→RU, with the following distribution: out of 148 sentence pairs, we have 141 *No Hallucination* (96.6%), 1 *Small* (0.68%), 2 *Partial* (1.4%), and 4 *Full* (2.8%). High class imbalance can affect the ability of model to perform well ([Prusa et al., 2016](#); [Sordo and Zeng, 2005](#); [Fernández et al., 2013](#)).

2.3 LLMs for hallucination detection

We assessed the performances of eight LLMs, mixing capabilities models across LLMs families. We evaluate OpenAI’s GPT4-turbo and GPT4o; Cohere’s Command R and Command R+; Mistral’s Mistral-8x22b; Anthropic’s Claude Sonnet and Claude Opus and Meta’s Llama3-70B.² More details about the selection are in [subsection D.2](#).

First, we built our prompt design by differentiated system and user prompts for better results ([Kong et al., 2024](#)). The system prompt contained the task description, and optionally, the inclusion of Chain-of-Thought (CoT), while the user prompt

contained, for each sentence pair, the source text and MT text, as well as a direct hallucination classification question.

We derived the task description prompts from the *Evaluate Hallucination* and *Evaluate Coherence in the Summarization Task* prompts in *G-Eval* ([Liu et al., 2023](#)). The CoT prompts were inspired by *Evaluation Steps* from *G-Eval*, and by the human annotation guidelines and severity level definitions from *HalOmi*. All prompts are available [Appendix C](#), with [Figure 3](#) showing one of the prompts used for binary detection. More details about the chosen hyperparameters with LLMs can be found in [Appendix D](#).

We determined the optimal prompts for each model using the DE↔EN validation set, evaluating three prompts and two CoT proposals for binary detection. The best prompt for each model was selected based on the average MCC across both translation directions. The MCC was chosen as the primary metric for binary detection due to its superiority in providing a single, easily interpretable value between -1 and +1. This value encapsulates the model’s performance for the confusion matrix scores, making it more robust to class imbalance.

2.4 Embeddings

We assessed the performance of three LLM-related embedding spaces: OpenAI’s text-embedding-3-large, Cohere’s Embed v3, and Mistral’s mistral-embed. Additionally, we included SONAR, the multilingual embedding space used as the base for BLASER-QE. Specifically, we calculated the cosine distance between embeddings of the source text and the machine-translated text. This approach draws on previous studies showing that hallucinated translations tend to have embeddings that are significantly distanced from those of the source text ([Dale et al., 2022](#)).

We binarised the cosine similarity scores of embeddings using an optimal threshold value determined from the validation set. This threshold, established by maximizing the F1-score from the precision-recall curve, was then applied to the test set for binary hallucination detection across all language pairs. Each embedding space was independently processed to maintain the integrity of the evaluation.

²GPT3.5, Mistral Large and Llama3-8B were initially taken into account, but were excluded due to poor task understanding.

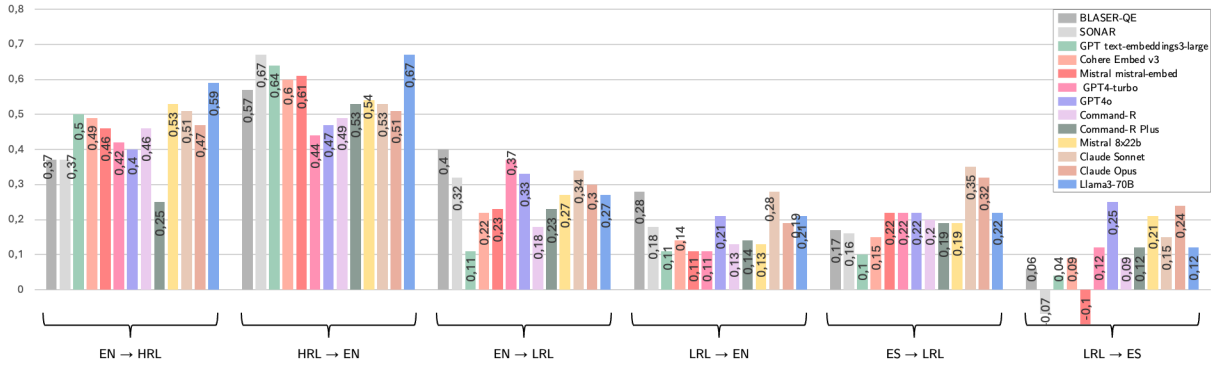


Figure 2: MCC average score across high and low resource levels, for different directions. The best performing models differ significantly between HRLs and LRLs. For HRLs, Llama3-70B greatly outperforms other methods, whereas for LRLs, best performers differ from and to LRLs, with Claude and GPT models closely competing. Embeddings demonstrate impressive results, particularly for the EN→HRL directions.

3 Results

LLMs are the new SOTA for hallucination detection The results in Figure 2 and Figure 15 demonstrate that LLMs have the best overall performance across languages for binary hallucination detection. Specifically, Llama3-70B surpasses the previous best performing model, BLASER-QE, by +5 points, with an MCC of 0.43. For HRLs, 10 out of 12 evaluated methods outperform BLASER-QE (0.46), with Llama3-70B greatly improving over the baseline by 16 points (0.63). Notably, the results show that the choice of LLM should rely on the resource level; as for LRLs, Claude Sonnet achieves the highest average MCC. However, GPT4o was the more robust LLM across all languages, with the lowest standard deviation. Finally, for 13 out of the 16 evaluated translation directions, the evaluated methods outperform BLASER-QE, with the exception of KS→EN, YO→EN and EN→MNI. Our findings on LLMs’ superior hallucination detection capabilities align with prior research on their effectiveness in MT quality assessment (Kocmi and Federmann, 2023).

Embedding-based hallucination detectors remain competitive for HRLs For HRLs, simple embedding-based methods display competitive capabilities, outperforming more sophisticated models in five out of eight translation directions. For instance, although BLASER-QE is a more advanced model based on SONAR, SONAR exhibits comparable or superior performances in most HRLs directions. This suggests that the effectiveness of these methods may be highly sensitive on their training data, and hence to the resource level, as we observe SOTA performances for HRLs and subop-

timal results for LRLs. Additionally, the embeddings’ performance may be highly dependent on the threshold chosen using the EN↔DE validation set, generalizing well for HRLs but not for LRLs.

LLMs’ contrastive performances across LRLs First, while Llama3-70B obtains the best performance overall, it was outperformed in most translation directions, especially in LRL. This result reveals a HRLs-centric approach of the model but also concludes that there is not one-LLM fits all resource levels. Secondly, for LRLs, models such as Sonnet, Opus, GPT4o, and Mistral —in order of decreasing performances, achieve higher scores, supporting the feasibility of employing LLMs in settings encompassing a wide range of languages. These results should be contrasted with a wide difference of hallucination distribution across resource levels, for example with the MN→EN direction which only has 28% *No hallucination* sentence pairs. More precisely, Sonnet and BLASER-QE perform on par for LRL, with the particularity that BLASER-QE has a significantly higher rate of false negatives, while Sonnet maintains a more balanced ratio of false positives to negatives. Moreover, BLASER-QE performs well in translations from English and comparably to Sonnet in translations to English, but falls short in non-English-centric translations, which follows the same trends as previously reported models in (Dale et al., 2023). Figure 2 and Figure 15 provide a more detailed view of these performance metrics.

Embeddings are high performers for non-Latin scripts, while LLMs can generalise to non-English centric translations For HRLs→EN directions with source scripts different than Latin

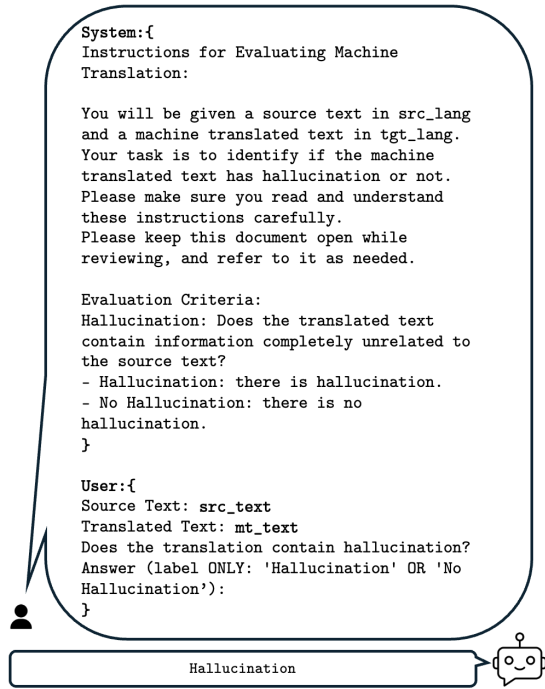


Figure 3: Binary detection prompt sample.

(AR, RU, ZH), embeddings are the best performers, suggesting high capabilities with cross-script transfer learning. These observations align with the findings of [Hada et al. \(2023\)](#), who report decreased performance for non-Latin scripts in LLM-based evaluators. In the two non-English centric translation directions (ES $\leftrightarrow$ YO), Opus outperforms by far both BLASER-QE (0.11) and the best embedding Mistral (0.12), with a score of 0.28. Unlike the overall LRLs trends, Opus outperforms Sonnet for this direction pair: this can suggest that the advanced analytical capabilities of LLMs can generate improved results even in scenarios with limited relevant training data. Remarkably, in the YO $\rightarrow$ ES translation direction, six out of our fourteen methods and BLASER-QE exhibit scores close to random guessing (within the $[-1, +1]$ range). This observation underscores the pressing need for enhanced capabilities in detecting hallucinations in non-English-centric translation settings. [Figure 1](#) presents two examples that highlight the challenges faced by LLMs when dealing with non-Latin scripts, with the exception of Llama3-70B. Additionally, it illustrates how embeddings may struggle with reasoning capabilities in non-English centric contexts.

4 Conclusion

In this work, we demonstrates that LLMs and embedding semantic similarity are highly effective

for hallucination detection in machine translation, with LLMs establishing a new state-of-the-art performance across both high and low-resource languages. Our findings suggest that the optimal model selection depends on specific contexts, such as resource level, script, and translation direction. Our study highlights the practical advantage of reference-free models like LLMs, which allow real-time hallucination detection without relying on external knowledge ([Su et al., 2024](#); [Zhang et al., 2023](#)). However, there remains a critical need for further research to improve hallucination detection, particularly in low-resource and non-English-centric translation settings.

Limitations

Despite the promising results obtained by LLMs and embedding-based methods in our evaluation, there are certain limitations that should be noted.

First, the dataset shows distribution imbalance across translation directions, with different trends for high and low resource languages, even after binarisation (see [Appendix A](#)): The HRLs show a pronounced data imbalance towards *No Hallucination* labels, with distribution between 79% and 94%. Moreover, for LRLs, there’s a broader interval, from 28% to 85%. This imbalance often results in models that classify translations as *No hallucination* being more frequently correct for HRLs than for LRLs, thereby introducing a bias into the binary evaluation. Moreover, the translation direction display a qualitative bias, as shown [subsection A.3](#): HRLs and LRLs don’t have the same selection distribution which display a potential bias towards hallucination. Future dataset improvements should prioritize larger, more diverse samples, non-Latin scripts, and non-English centric translations. Using consistent source text across languages and balancing hallucination severity levels would enable more sophisticated methods, improve generalizability, and allow for a fair evaluation of models’ hallucination detection capabilities.

It is important to note there is a possibility of test set contamination, which is a common challenge in LLM research when the full training data is not publicly available. This issue primarily affects HRLs, where LLMs are predominantly trained; therefore, the impact on LRLs performance is expected to be minimal.

The validation set used to identify the optimal threshold for non-LLM methods and the best prompt for LLMs only included EN↔DE translations. To improve parameter optimization and generalization across various translation directions, especially for LRLs, cross-validation is recommended for future research, as suggested by [Dale et al. \(2023\)](#) and initially planned for our study. However, financial constraints associated with benchmarking non-open source models prevented the implementation of this approach. Future work should focus on developing novel approaches that excel on well-studied HRLs while generalizing effectively to LRLs, assessing robustness, or exploring alternative methods to address this challenge within the limitations of dataset size.

Finally, for benchmarking purposes, only the

previous state-of-the-art was included for comparison against the newly evaluated methods. Therefore, for a more comprehensive analysis, it is recommended to include additional methods previously evaluated by *HalOmi*. Moreover, the benchmark can be further strengthened by identifying fine-grained hallucination spans to enhance interpretability.

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A Dataset description

A.1 Language acronyms mapping

The languages acronyms follow this mapping throughout the paper: Arabic (AR), Chinese (ZH), English (EN), German (DE), Kashmiri (KA), Manipuri (MN), Russian (RU), Spanish (ES), and Yoruba (YO).

A.2 Hallucination distribution

A.2.1 Distribution of Hallucination in the severity ranking framework

Direction	Total	1 No	2 Small	3 Partial	4 Full
DE→EN	155	140 90.32%	2 1.29%	2 1.29%	11 7.10%
EN→DE	146	132 90.41%	3 2.05%	2 1.37%	9 6.16%
Total	301	272 68.25%	5 1.25%	4 1.00%	20 5.01%

Direction	Total	1 No	2 Small	3 Partial	4 Full
EN→AR	144	136 94.44%	2 1.39%	2 1.39%	4 2.78%
AR→EN	156	132 84.62%	5 3.21%	2 1.28%	17 10.90%
EN→RU	146	141 96.58%	1 0.68%	2 1.37%	2 1.37%
RU→EN	158	146 92.41%	3 1.90%	2 1.27%	7 4.43%
EN→ES	153	131 85.62%	8 5.23%	3 1.96%	11 7.19%
ES→EN	160	127 79.38%	17 10.63%	4 2.50%	12 7.50%
EN→ZH	160	131 81.88%	5 3.13%	4 2.50%	20 12.50%
ZH→EN	159	127 79.87%	9 5.66%	7 4.40%	16 10.06%
EN→KA	184	111 60.33%	8 4.35%	30 16.30%	35 19.02%
KA→EN	151	89 58.94%	15 9.93%	32 21.19%	15 9.93%
EN→YO	195	166 85.13%	4 2.05%	11 5.64%	14 7.18%
YO→EN	146	124 84.93%	4 2.74%	10 6.85%	8 5.48%
EN→MN	197	78 39.59%	52 26.40%	54 27.41%	13 6.60%
MN→EN	152	43 28.29%	45 29.61%	58 38.16%	6 3.95%
ES→YO	151	97 64.24%	16 10.60%	29 19.21%	9 5.96%
YO→ES	152	80 52.63%	26 17.11%	37 24.34%	9 5.92%
Total	2564	1859 72.47%	220 8.58%	287 11.19%	198 7.72%

Table 1: Although fine-grained severity ranking is advantageous for most applications, the rarity of occurrences within each hallucination category may lead to results that lack significance and generalizability due to constrained sample sizes. Notably, within the *HalOmi* dataset, 11 of the 18 language directions include fewer than five samples in at least one hallucination category. To address this limitation, we propose a shift toward binary hallucination detection, where all instances of hallucinations are classified as such, irrespective of their severity. This approach enhances the robustness of the analysis and the significance of results while still evaluating the model’s ability to separate even *Small hallucination* (one word in a sentence) from *No hallucinations*.

A.2.2 Distribution of Hallucination in the binary detection framework

Direction	Total	0 No Hallucination	1 Hallucination
DEU→EN	155	140 90.32%	15 9.68%
EN→DE	146	132 90.41%	14 10.00%
Total	301	272 68.25%	29 31.75%

Table 2: Validation set distribution for binary detection, across translation directions, for HRLs and LRLS

Direction	Total	0 No Hallucination	1 Hallucination
EN→AR	144	136 94.44%	8 5.56%
AR→EN	156	132 84.62%	24 15.38%
EN→RU	146	141 96.58%	5 3.42%
RU→EN	158	146 92.41%	12 7.59%
EN→ES	153	131 85.62%	22 14.38%
ES→EN	160	127 79.38%	33 20.63%
EN→ZH	160	131 81.88%	29 18.13%
ZH→EN	159	127 79.87%	32 20.13%
EN→KA	184	111 60.33%	73 39.67%
KA→EN	151	89 58.94%	62 41.06%
EN→YO	195	166 85.13%	29 14.87%
YO→EN	146	124 84.93%	22 15.07%
EN→MN	197	78 39.59%	119 60.41%
MN→EN	152	43 28.29%	109 71.71%
ES→YO	151	97 64.24%	54 35.76%
YO→ES	152	80 52.63%	72 47.37%
Total	2564	1859 72.47%	705 27.53%

Table 3: Testing set distribution for binary detection, across translation directions, for HRLs and LRLS

A.3 Selection distribution

The selection information from the *HalOmi* dataset indicates the sampling strategy used to select sentence pairs for each translation direction and data source, which includes *uniform* sampling to maintain data diversity, *biased* sampling favoring potentially problematic translations based on detector quantiles, and *worst* sampling, according to the detectors to increase the likelihood of capturing hallucinations. A closer look at the selection distribution is available [Figure 4](#)

B Ablation study

The ablation study focus on hallucination **severity ranking**. We present results for comparability with

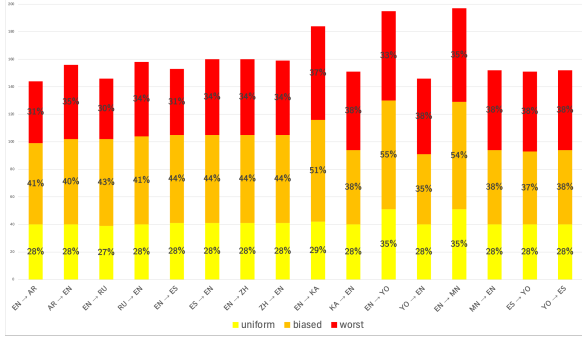


Figure 4: **Selection type distribution** This graph shows that the three EN→LRLs not only have more sentences, but also have way more *biased* sentences than other directions, which suggests a higher propensity to hallucinate.

Dale et al. (2023), which assesses the methods’ abilities to accurately rank hallucinations by severity (e.g., full hallucinations ranked higher than partial ones, and any hallucinations ranked above non-hallucinations). The employed metric is an adaptation of the ROC AUC for multiclass tasks, which calculates the percentage of incorrectly ranked sentence pairs with different labels and subtracts this value from the perfect score of 1. We compute these metrics separately for each translation direction to assess the detector’s performance across different language pairs.

B.1 LLMs for severity ranking

B.1.1 Prompt design

We designed tailored prompts for this approach, just as for our main binary approach, this time to generate multiclass predictions. For severity ranking, each prompt has a different assigned CoT.

Severity ranking, Prompt1: *G-Eval* inspired
See Figure 5

Severity ranking, Prompt2: *G-Eval* inspired, with *HalOmi*’s hallucination definition See Figure 6

Severity ranking, Prompt3: *G-Eval* inspired, with *HalOmi*’s hallucination definition, and language precision See Figure 7

Chain of Thoughts for severity ranking See Figure 8 and Figure 9

B.1.2 Prompt evaluation

We evaluated three prompts and two CoT variations on the validation set to select the best prompt (Table 4). The prompt that achieved the highest average ROC AUC for both directions (DE↔EN)

System:

Human Evaluation of Machine Translation Systems:

Hallucination Evaluation Criteria: Does the translated text contain information unrelated to the source text?

- **No hallucination:** The translated text does not contain any hallucinated words.
- **Small hallucination:** The translated text contains 1-2 hallucinated words.
- **Partial hallucination:** The translated text includes at least 3 hallucinated words, but not all words are hallucinated.
- **Full hallucination:** Nearly all words in the translated text are hallucinated, with the exception of perhaps 1-2 words.

Note: The labels are mutually exclusive; for example, a translation with a partial hallucination does not qualify as a full hallucination.:

User:

Source Text: src_text

Translated Text: mt_text

Respond with the hallucination level label ONLY. Answer (Labels=[‘No hallucination’, ‘Small hallucination’, ‘Partial hallucination’, ‘Full hallucination’])

Figure 5: **Severity Ranking Prompt 1** - from *G-Eval*

was chosen for each method. Subsequently, in the testing phase, each model was assessed with its optimal prompt.

B.2 Embeddings for severity ranking

We computed the cosine similarity between the source text and machine-translated text embeddings for each embedding space and took the negative of these results. This approach ensures that hallucinations (indicative of embeddings that are farther apart) correspond to higher numbers, consistent with the ranking scale used in hallucination evaluation. Since this method does not require parameter tuning, the validation set was not utilized for thresholding in contrast to the binary approach.

B.3 Results

In the same way as in the binary detection setting, the validation results Table 4 allowed to se-

lect the optimal prompt for each LLM, and then evaluate this best prompt across the test set, using here the ROC AUC score. Testing results are displayed [Table 5](#), and presents ROC AUC scores for all methods per translation direction. For HRLs, embeddings' high performance remains consistent with the binary hallucination approach. However, BLASER-QE remains the state-of-the-art in overall performance for severity ranking. The generalizability of these results requires further evaluation due to significant class imbalances in the dataset. Notably, in 11 of the 18 language directions, fewer than five samples are present in at least one hallucination severity category, see [Appendix A](#).

C Prompts

We used two types of CoTs: One based on the human guidelines for hallucination detection, and the other based on the severity level definition, that was readapted to each case. For **binary detection**, two CoTs were tested for three prompts.

Binary detection, Prompt1 - from *G-Eval* See [Figure 10](#)

Binary detection, Prompt2 - from *G-Eval* with language precision See [Figure 11](#)

Binary detection, Prompt3 - Human designed prompt See [Figure 12](#)

Binary detection, Chain of Thoughts See [Figure 13](#) and [Figure 14](#)

D LLMs experiments

D.1 LLMs hyperparameters

For the evaluation of LLMs, we used *LangChain* to ensure reproducibility of results, except for Llama3-70B that was ran locally. We set the TEMPERATURE to 0 for minimum randomness and the MAX_OUTPUT_TOKEN to 15 to avoid verbose. All the experiments were zero-shot, with an exhaustive label (for example, [*'Hallucination'*, *'No Hallucination'*] for **binary detection**). These choices showed the highest performances in previous research ([Kocmi and Federmann, 2023](#)) ([Wei et al., 2022](#)).

D.2 LLMs selection

We selected the following models for our evaluation: GPT4-turbo, widely adopted in both academic research and industrial applications due to its robust performance and versatility; GPT4o, the

latest GPT model, optimised for better human-computer interaction; Command-R, known for its large context window, well-suited for tasks that require extended language understanding and generation; Command R+, an enhanced version of Command R, demonstrating strong performance in multilingual tasks, achieving impressive BLEU scores in benchmark datasets such as [FLoRES](#) and [WMT23](#); Mistral 8x22b, currently the most performant open model from Mistral, excelling in various language tasks; Claude Sonnet, showing strong capabilities in multilingual tasks, similar to Command R+; Claude Opus, known as the "most intelligent" Claude model, offering advanced language understanding and generation capabilities; and Llama3-70B, the most capable openly available LLM from Meta, evaluated in its 70B size for comprehensive performance analysis. These models were chosen based on their demonstrated performance in various benchmarks and their potential to handle a wide range of language tasks effectively.

Model	Prompt1		Prompt2		Prompt3		AVG	
	no CoT	CoT1	no CoT	CoT2	no CoT	CoT2	Mean	Std.
GPT4-Turbo	0.78	0.70	0.83	0.81	0.82	0.81	0.79	0.05
GPT4o	0.81	0.82	0.83	0.83	0.83	0.83	0.83	0.01
Command R	0.82	0.79	0.77	0.80	0.83	0.79	0.80	0.02
Command R+	0.75	0.76	0.79	0.80	0.77	0.75	0.77	0.02
Mistral 8x22b	0.57	0.58	0.77	0.67	0.69	0.67	0.66	0.07
Sonnet	0.82	0.83	0.82	0.79	0.81	0.83	0.82	0.01
Opus	0.79	0.80	0.82	0.83	0.85	0.77	0.81	0.03
Llama3-70B	0.80	0.81	0.81	0.78	0.81	0.78	0.80	0.01

Table 4: Validation results for hallucination detection across prompt variations for severity ranking.

Model	EN→HRL				HRL→EN				EN→LRL			LRL→EN			ES→YO		YO→ES		AVG		
	AR	RU	ES	ZH	AR	RU	ES	ZH	KA	YO	MN	KA	YO	MN					HRL	LRL	Overall
GPT text-embedding-3-large	0.89	0.82	0.84	0.92	0.91	0.94	0.87	0.87	0.71	0.7	0.54	0.56	0.68	0.6	0.62	0.51			0.88	0.62	0.75
Cohere Embed v3	0.84	0.87	0.83	0.88	0.9	0.96	0.89	0.83	0.75	0.73	0.54	0.58	0.74	0.64	0.65	0.59			0.88	0.65	0.76
Mistral-embed	0.92	0.88	0.82	0.85	0.92	0.86	0.86	0.83	0.72	0.7	0.56	0.53	0.68	0.61	0.63	0.53			0.87	0.62	0.74
SONAR	0.89	0.79	0.85	0.77	0.93	0.93	0.85	0.87	0.81	0.8	0.69	0.73	0.79	0.73	0.69	0.62			0.86	0.73	0.8
GPT4-Turbo	0.8	0.72	0.65	0.8	0.86	0.91	0.86	0.79	0.61	0.57	0.26	0.47	0.43	0.31	0.38	0.4			0.8	0.43	0.61
GPT4o	0.71	0.74	0.65	0.8	0.86	0.86	0.74	0.8	0.64	0.58	0.3	0.47	0.59	0.4	0.45	0.41			0.77	0.48	0.63
Command R	0.56	0.88	0.61	0.83	0.86	0.84	0.77	0.68	0.47	0.51	0.19	0.16	0.19	0.33	0.37	0.3			0.75	0.32	0.53
Command R+	0.59	0.56	0.65	0.7	0.91	0.91	0.76	0.74	0.34	0.39	0.04	0.41	0.43	0.26	0.15	0.4			0.73	0.3	0.51
Mistral 8x22b	0.25	0.59	0.53	0.67	0.84	0.94	0.74	0.77	0.51	0.4	0.08	0.46	0.52	0.5	0.33	0.46			0.67	0.41	0.54
Sonnet	0.7	0.75	0.61	0.8	0.84	0.89	0.7	0.69	0.64	0.62	0.41	0.56	0.58	0.55	0.53	0.47			0.75	0.55	0.65
Opus	0.6	0.91	0.69	0.83	0.88	0.9	0.83	0.76	0.66	0.54	0.2	0.49	0.7	0.53	0.33	0.49			0.8	0.49	0.65
Llama3-70B	0.6	0.91	0.69	0.83	0.88	0.9	0.83	0.76	0.66	0.54	0.2	0.49	0.7	0.53	0.33	0.49			0.8	0.49	0.65
BLASER 2.0-QE	0.9	0.89	0.85	0.78	0.94	0.92	0.87	0.87	0.81	0.83	0.79	0.73	0.78	0.8	0.68	0.58			0.88	0.75	0.81

Table 5: ROC-AUC results for severity hallucination ranking across HRL and LRL directions.

Bold values indicate the best performing prompt per model.

System:

You will be given a 'Source Text' and its translation, the 'Translated Text'. Your task is to identify the hallucination level of the translated text.

Definition of Hallucination:

A word in the translated text is considered a hallucination if it introduces information that is completely unrelated to the source text.

Hallucination labels:

- **No hallucination:** The translated text does not contain any hallucinated words.
- **Small hallucination:** The translated text contains 1-2 hallucinated words.
- **Partial hallucination:** The translated text includes at least 3 hallucinated words, but not all words are hallucinated.
- **Full hallucination:** Nearly all words in the translated text are hallucinated, with the exception of perhaps 1-2 words.

Note: The labels are mutually exclusive; for example, a translation with a partial hallucination does not qualify as a full hallucination.:

User:

Source Text: src_text

Translated Text: mt_text

Provide exactly one of the following hallucination level labels as your response. Do not include any additional text or explanation:

- No hallucination
- Small hallucination
- Partial hallucination
- Full hallucination

Figure 6: **Severity Ranking Prompt 2** - from *G-Eval* with Hallucination definition

System:

You will be given a 'Source Text' in src_lang and its translation in tgt_lang, the 'Translated Text'. Your task is to identify the hallucination level of the translated text. Please make sure you read and understand these instructions carefully. Please keep this document open while reviewing, and refer to it as needed.

Definition of Hallucination:

A word in the translated text is considered a hallucination if it introduces information that is completely unrelated to the source text.

Hallucination labels:

- **No hallucination:** The translated text does not contain any hallucinated words.
- **Small hallucination:** The translated text contains 1-2 hallucinated words.
- **Partial hallucination:** The translated text includes at least 3 hallucinated words, but not all words are hallucinated.
- **Full hallucination:** Nearly all words in the translated text are hallucinated, with the exception of perhaps 1-2 words.

Note: The labels are mutually exclusive; for example, a translation with a partial hallucination does not qualify as a full hallucination.:

User:

Source Text: src_text

Translated Text: mt_text

Provide exactly one of the following hallucination level labels as your response. Do not include any additional text or explanation:

- No hallucination
- Small hallucination
- Partial hallucination
- Full hallucination

Figure 7: **Severity Ranking Prompt 3** - from *G-Eval* with Hallucination definition and language precision

Evaluation Steps:

1. Read the source text and the translated text carefully.
2. To decide whether the translated text contains hallucinations check if the source word “corresponds” to erroneous target tokens. For each word answer:
 - Does this source word fall into the common meaning category as this target word?
 - Does this source word have a semantic connection with this target word?
 - Can you try to come up with a reasonable theory on how this source word is associated with this target word?
 - If “no” to all the questions above, then hallucination. Keep a count of the number of hallucinated words for each sentence pair.
3. After reading all the source and translated text, assign a label to the pair based on the number of hallucinated words.

Figure 8: **Severity Ranking CoT 1** - from *HalOmi*'s human guidelines

Evaluation Steps:

1. Read the source text and the translated text carefully.
2. Initialize a counter ‘ $n = 0$ ’ for the number of hallucinated words.
3. For each word in the translated text, perform the following checks to determine if it is a hallucinated word:
 - Does this source word fall into the common meaning category as this target word?
 - Does this source word have a semantic connection with this target word?
 - Can you try to come up with a reasonable theory on how this source word is associated with this target word?
 - If “no” to all the questions above, then it is considered a hallucination. Increment ‘ n ’ by 1.
4. After analyzing each word in the translated text:
 - If ‘ $n == 0$ ’, assign the label ‘No hallucination’.
 - If ‘ n ’ is 1 or 2, assign the label ‘Small hallucination’.
 - If ‘ n ’ is 3 or more but not all words are hallucinated, assign the label ‘Partial hallucination’.
 - If nearly all words are hallucinated, assign the label ‘Full hallucination’.

Figure 9: **Severity Ranking CoT 2** - counting the number of hallucinated words

System:
Human Evaluation of Machine Translation Systems:
Hallucination Evaluation Criteria: Does the translated text contain information completely unrelated to the source text?
- *Hallucination*: there is hallucination.
- *No Hallucination*: there is no hallucination.

User:
Source Text: src_text
Translated Text: mt_text
Does the translation contain hallucination?
Answer (label ONLY: 'Hallucination' OR 'No Hallucination'):

Figure 10: **Binary detection Prompt 1** - from *G-Eval*

System:
Instructions for Evaluating Machine Translation:
You will be given a source text in src_lang and a machine translated text in tgt_lang. Your task is to identify if the machine translated text has hallucination or not. Please make sure you read and understand these instructions carefully. Please keep this document open while reviewing, and refer to it as needed.
Evaluation Criteria:
Hallucination: Does the translated text contain information completely unrelated to the source text?
- *Hallucination*: there is hallucination.
- *No Hallucination*: there is no hallucination.

User:
Source Text: src_text
Translated Text: mt_text
Does the translation contain hallucination?
Answer (label ONLY: 'Hallucination' OR 'No Hallucination'):

Figure 11: **Binary detection Prompt 2** - from *G-Eval* with language precision

System:
Instructions for Evaluating Machine Translation:
You will be given a source text in src_lang and a machine translated text in mt_lang. Your task is to identify if the machine translated text has hallucination or not. Please make sure you read and understand these instructions carefully. Please keep this document open while reviewing, and refer to it as needed.
Definition of Hallucination: The translated text is considered a hallucination if it introduces information that is completely unrelated to the source text.
Hallucination labels:

- **Hallucination:** there is hallucination.
- **No hallucination:** there is no hallucination.

User:
Source Text: src_text
Translated Text: mt_text
Provide exactly one of the following hallucination labels as your response. Do not include any additional text or explanation:

- *Hallucination*
- *No hallucination:*

Figure 12: **Binary detection Prompt 3** - Human designed prompt

Evaluation Steps:

1. Read the source text and the translated text carefully.
2. To decide whether the translated text contains hallucinations check if the source tokens "correspond" to erroneous target tokens. For each token answer:

- Does this source word fall into the common meaning category as this target word?
- Does this source word have a semantic connection with this target word?
- Can you try to come up with a reasonable theory on how this source word is associated with this target word?

3. If "no" to all the questions above, then hallucination

Figure 13: **Binary detection - CoT1:** from *HalOmi*'s human guidelines

Evaluation Steps:

1. Read the source text and the translated text carefully.
2. Initialize a counter 'n = 0' for the number of hallucinated words.
3. To decide whether the translated text contains hallucinations check if the source tokens "correspond" to erroneous target tokens. For each token answer:

- Does this source word fall into the common meaning category as this target word?
- Does this source word have a semantic connection with this target word?
- Can you try to come up with a reasonable theory on how this source word is associated with this target word?
- If "no" to all the questions above, then hallucination

4. After analyzing each word in the translated text:

- If 'n == 0', assign the label 'No hallucination'.
- If 'n' is 1 or more, assign the label 'Hallucination'."

Figure 14: **Binary detection - CoT2:** from *HalOmi*'s human guidelines and counting strategy

E Binary detection results

E.1 Validation results

Table 6 provides MCC scores per LLM for each of the prompts and CoT variations evaluated on the validation set. The most robust LLMs across prompt variations in the validation set, specifically Sonnet, GPT4o, and Llama3-70B, exhibit superior performance across language resource settings in the test set. This suggests that extensive prompt engineering might not be required for these models in the current task, as the performance using the optimal prompt from the validation set aligns with high performance on the test set.

E.2 Test results

Figure 15 and Figure 16 display the performances of the evaluated methods on the test set, grouped by translation directions. To account for the significant class imbalance, multiple metrics are employed to ensure a more comprehensive and unbiased analysis. These include the MCC, binary F1-score, macro F1-score, and precision-recall area under the curve (AUC). The results indicate that the highest scores for HRLs are achieved in translations to English, whereas for LRLs, the highest scores are from translations originating in English or Spanish. Additionally, these findings underscore that no single model uniformly excels across all translation directions. Finally, the model rankings remain consistent across metrics in HRLs settings. However, there is greater variability in LRL scenarios, particularly for non-English centric translation directions. This variability is largely due to models like BLASER-QE and SONAR, exhibiting a high ratio of non-hallucination predictions, while others, such as Command-R and GPT-embeddings, show a stronger tendency towards hallucination predictions.

Model	Prompt1		Prompt2		Prompt3			AVG	
	no CoT	CoT1	no CoT	CoT1	no CoT	CoT1	CoT2	Mean	Std.
Binary Detection (MCC)									
GPT4-Turbo	0.53	0.55	0.55	0.50	0.45	0.51	0.47	0.51	0.04
GPT4o	0.44	0.44	0.51	0.45	0.44	0.47	0.48	0.46	0.03
Command R	0.43	0.37	0.54	0.47	0.51	0.53	0.55	0.49	0.07
Command R+	0.72	0.72	0.57	0.69	0.54	0.72	0.64	0.66	0.08
Mistral 8x22b	0.51	0.57	0.52	0.61	0.69	0.65	0.69	0.61	0.07
Sonnet	0.67	0.68	0.69	0.68	0.68	0.69	0.68	0.68	0.01
Opus	0.57	0.50	0.53	0.56	0.73	0.64	0.59	0.59	0.08
Llama3-70B	0.74	0.76	0.74	0.72	0.81	0.79	0.79	0.76	0.03

Table 6: Validation results for binary hallucination detection across prompt variations. Bold values indicate the best performing prompt per model. In the case of ties, we favor shorter prompts without CoT.

		Performance per Language Direction and Detection Method															
		MCC								F1-Binary							
EN→AR		0.41	0.40	0.35	0.40	0.29	0.27	0.38	0.43	0.14	0.25	0.39	0.39	0.39	-0.44	0.43	0.38
AR→EN		0.69	0.76	0.77	0.69	0.70	0.47	0.52	0.56	0.57	0.62	0.58	0.59	0.77	-0.74	0.78	0.80
EN→RU		0.13	0.17	0.20	0.20	0.24	0.27	0.34	0.34	-0.03	0.59	0.52	0.26	0.53	-0.17	0.20	0.22
RU→EN		0.64	0.75	0.61	0.67	0.67	0.34	0.43	0.39	0.50	0.53	0.43	0.48	0.72	-0.67	0.74	0.64
EN→ES		0.61	0.41	0.66	0.66	0.70	0.61	0.53	0.52	0.55	0.55	0.59	0.59	0.66	-0.65	0.32	0.67
ES→EN		0.51	0.50	0.58	0.51	0.47	0.55	0.50	0.59	0.63	0.55	0.64	0.58	0.68	-0.57	0.49	0.57
EN→ZH		0.31	0.51	0.78	0.71	0.63	0.53	0.35	0.57	0.36	0.72	0.56	0.63	0.78	-0.44	0.60	0.81
ZH→EN		0.45	0.65	0.60	0.55	0.60	0.39	0.41	0.43	0.41	0.45	0.46	0.39	0.51	-0.57	0.68	0.61
EN→KA		0.39	0.44	0.19	0.42	0.39	0.53	0.47	0.30	0.34	0.49	0.49	0.52	0.39	-0.61	0.52	0.57
KA→EN		0.37	0.29	0.14	0.02	0.06	0.06	0.12	0.14	0.03	0.15	0.36	0.21	0.21	-0.48	0.47	0.58
EN→YO		0.43	0.31	0.21	0.25	0.26	0.37	0.38	0.25	0.35	0.26	0.47	0.36	0.33	-0.51	0.38	0.32
YO→EN		0.40	0.11	0.17	0.20	0.17	0.16	0.22	0.06	0.20	0.15	0.25	0.20	0.26	-0.47	0.14	0.31
EN→MN		0.37	0.21	-0.07	0.00	0.03	0.20	0.14	-0.01	0.00	0.06	0.07	0.02	0.11	-0.47	0.27	0.74
MN→EN		0.07	0.15	0.02	0.19	0.12	0.12	0.30	0.19	0.18	0.09	0.21	0.16	0.16	-0.04	0.14	0.83
ES→YO		0.17	0.16	0.10	0.15	0.22	0.22	0.22	0.20	0.19	0.19	0.35	0.32	0.22	-0.24	0.19	0.52
YO→ES		0.06	-0.07	0.04	0.09	-0.10	0.12	0.25	0.09	0.12	0.21	0.15	0.24	0.12	-0.30	0.03	0.59
		F1-Macro								Precision-Recall AUC							
EN→AR		0.70	0.70	0.67	0.70	0.64	0.59	0.65	0.66	0.57	0.62	0.69	0.67	0.67	-0.46	0.45	0.42
AR→EN		0.84	0.87	0.88	0.84	0.85	0.67	0.72	0.76	0.78	0.80	0.77	0.79	0.88	-0.76	0.83	0.82
EN→RU		0.57	0.59	0.60	0.60	0.61	0.56	0.57	0.61	0.48	0.79	0.74	0.62	0.76	-0.19	0.21	0.24
RU→EN		0.82	0.86	0.80	0.84	0.84	0.60	0.65	0.64	0.74	0.74	0.67	0.72	0.86	-0.68	0.81	0.65
EN→ES		0.80	0.63	0.81	0.80	0.83	0.81	0.76	0.76	0.74	0.74	0.78	0.77	0.80	-0.69	0.65	0.75
ES→EN		0.74	0.70	0.75	0.69	0.70	0.77	0.75	0.79	0.78	0.77	0.82	0.78	0.81	-0.66	0.69	0.76
EN→ZH		0.58	0.75	0.89	0.84	0.80	0.73	0.59	0.75	0.65	0.86	0.78	0.81	0.89	-0.58	0.63	0.84
ZH→EN		0.73	0.81	0.77	0.74	0.77	0.63	0.65	0.68	0.71	0.71	0.71	0.66	0.75	-0.61	0.77	0.75
EN→KA		0.69	0.66	0.57	0.66	0.63	0.75	0.73	0.60	0.59	0.71	0.74	0.72	0.61	-0.70	0.75	0.66
KA→EN		0.63	0.61	0.52	0.51	0.45	0.34	0.50	0.52	0.40	0.57	0.66	0.61	0.57	-0.71	0.65	0.67
EN→YO		0.71	0.65	0.41	0.61	0.54	0.68	0.68	0.47	0.59	0.60	0.73	0.67	0.63	-0.55	0.45	0.57
YO→EN		0.70	0.53	0.41	0.58	0.54	0.28	0.50	0.30	0.57	0.45	0.48	0.57	0.57	-0.53	0.28	0.54
EN→MN		0.56	0.44	0.38	0.50	0.50	0.59	0.44	0.43	0.28	0.50	0.42	0.34	0.53	-0.85	0.79	0.79
MN→EN		0.24	0.30	0.44	0.59	0.49	0.46	0.65	0.59	0.36	0.53	0.60	0.48	0.52	-0.86	0.87	0.86
ES→YO		0.51	0.49	0.47	0.56	0.61	0.59	0.57	0.58	0.45	0.45	0.60	0.53	0.47	-0.53	0.55	0.63
YO→ES		0.48	0.35	0.49	0.52	0.43	0.47	0.61	0.48	0.37	0.60	0.58	0.54	0.55	-0.56	0.37	0.67
BLASER-QE																	
SONAR																	
GPT text-embeddings3-large																	
Cohere Embed v3																	
Mistral mistral-embed																	
GPT4-turbo																	
GPT4o																	
Command-R																	
Command-R Plus																	
Mistral 8x22b																	
Claude Sonnet																	
Claude Opus																	
Llama3-70B																	
BLASER-QE																	
SONAR																	
GPT text-embeddings3-large																	
Cohere Embed v3																	
Mistral mistral-embed																	
GPT4-turbo																	
GPT4o																	
Command-R																	
Command-R Plus																	
Mistral 8x22b																	
Claude Sonnet																	
Claude Opus																	
Llama3-70B																	

Figure 15: MCC, binary F1, macro F1 and precision-recall AUC scores for hallucination binary detection across 16 translation directions per method.

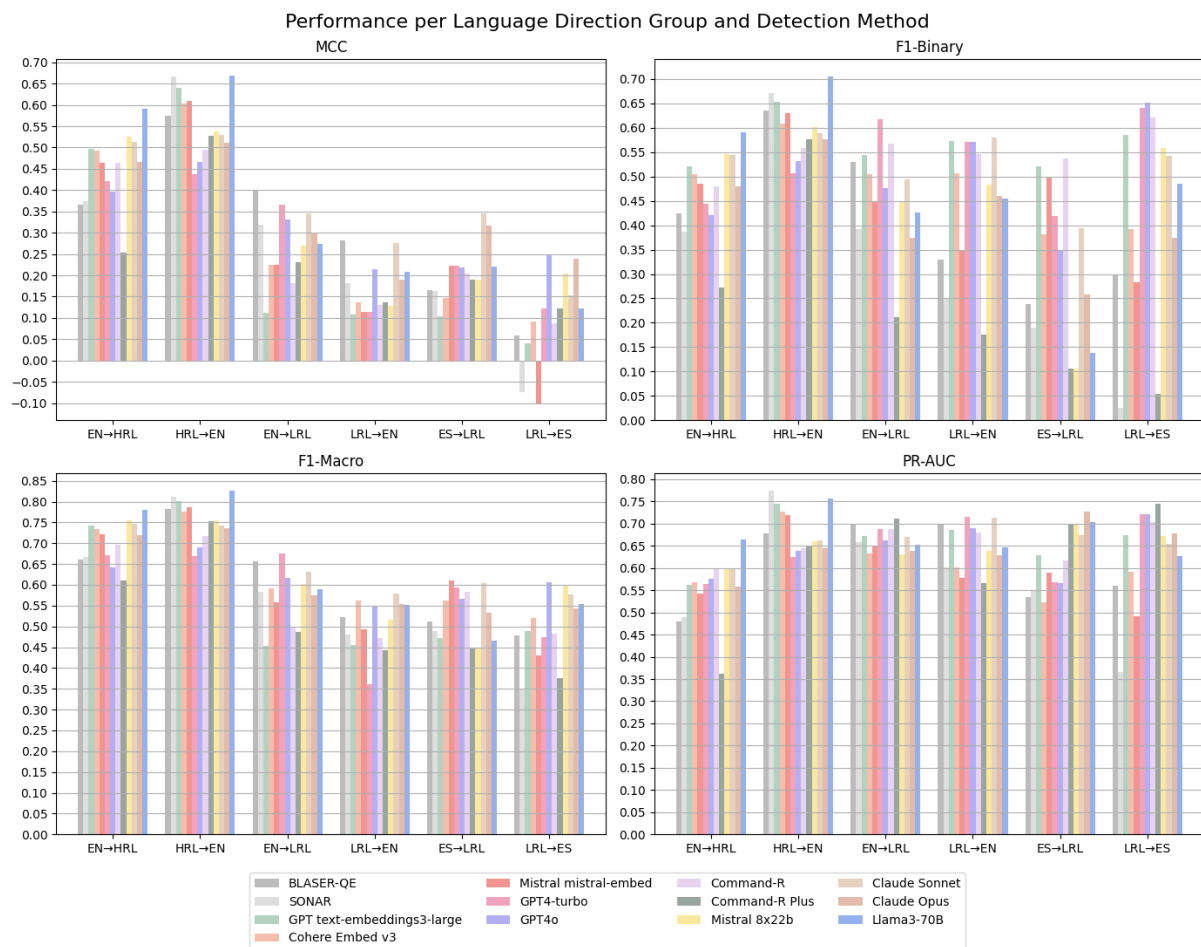


Figure 16: MCC, binary F1, macro F1 and precision-recall AUC average scores across high and low resource levels, for different directions.